Measuring Policy Ambiguity's Causal Impact on Agent Safety: A Matched-Pair Framework with Judge Calibration

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Abstract

LLM-based agents increasingly operate under natural-language policy specifications that inevitably contain ambiguity. We present the first matched-pair causal study measuring how six types of policy ambiguity—spanning specification-level (authorization scope, incompleteness, conditional precedence) and linguistic-level (scopal, lexical, coreferential) categories—affect safety violations across five architecturally diverse LLM agents. Using 300 ambiguous/unambiguous clause pairs injected into the τ^2-bench environment, we collect 2,997 agent episodes and apply a two-stage evaluation pipeline with LLM-as-judge assessment followed by human-anchored calibration. Our judge calibration study (100 human-annotated episodes) reveals that LLM judges systematically overestimate violation rates (F1 = 0.64, $\kappa = 0.56$ vs. humans) despite excellent inter-judge agreement ($\kappa = 0.91$) and test-retest reliability (ICC = 0.919). After correction, ambiguous policies still cause a substantial +19.5 percentage point increase in violations (OR = 6.2, $p < 0.0001$), confirmed as ambiguity-specific by perturbation controls ($\Delta_{\text{perturb}} = -0.67$pp). We find that ambiguity types differ in severity ($\chi^2 = 16.4$, $p = 0.006$) and that models exhibit distinct vulnerability profiles (model×type interaction, $p = 0.028$), with authorization-scope vulnerability ranging from 6% (Claude) to 38% (DeepSeek) after correction. Qualitative analysis of 614 violation traces reveals pervasive *ambiguity blindness* (4.6% awareness rate, $\kappa = 0.71$ vs. human anchor). Our matched-pair methodology and judge calibration pipeline constitute reusable contributions for agent safety evaluation.

1 Introduction

Consider a customer-service agent operating under a return policy that states *"items may be returned within 30 days."* The scopal ambiguity of "items"—does it include customized products?—leads the agent to grant a full refund on a $200 personalized order. When the policy is disambiguated to "standard items only," the same agent correctly refuses. A single word of ambiguity causes an unauthorized financial action.

As LLM-based agents are deployed with natural-language policy specifications (Yao et al., 2023; Schick et al., 2024), the safety implications of policy ambiguity become urgent. Prior work classifies ambiguity types in requirements (Berry and Kamsties, 2003; Massey et al., 2014; Li et al., 2024a) and evaluates agent safety under adversarial or missing policies (Ruan et al., 2024; Xie et al., 2024), but no study causally isolates ambiguity's effect on safety violations at type-level granularity across diverse model families.

We address this gap with three contributions:

(1) Matched-pair causal framework. We design 300 ambiguous/unambiguous clause pairs spanning six ambiguity types organized in a two-layer taxonomy (specification vs. linguistic), inject them into τ^2-bench (τ^2-bench, 2024) scenarios, and collect 2,997 episodes across five models (GPT-5.4, GPT-4.1, Claude Sonnet 4.6, Qwen3-235B, DeepSeek-V3). Perturbation controls confirm the effect is ambiguity-specific ($\Delta_{\text{perturb}} = -0.67$pp vs. corrected ambiguity $\Delta = +19.5$pp).

(2) Judge calibration pipeline. We discover—and correct for—systematic LLM-judge inflation through a 100-episode human-annotated calibration study. Despite excellent inter-judge reliability ($\kappa = 0.91$, ICC = 0.919), judges achieve only F1 = 0.64 against human ground truth ($\kappa = 0.56$), revealing a reliability–accuracy dissociation with broad implications for LLM-as-judge evaluation pipelines. After correction, the headline effect decreases from +34.0pp to +19.5pp (OR from 15.1 to 6.2) but remains highly significant ($p < 0.0001$).

(3) Corrected differential analysis. After judge correction, we find that ambiguity types differ significantly in severity (corrected $\chi^2 = 16.4$, $p = 0.006$, $V = 0.112$), with incompleteness most dangerous (38%) and coreferential least (16%). A model×type interaction ($p = 0.028$) reveals that authorization-scope vulnerability varies 32pp across models (6%–38%), while incompleteness remains universally dangerous (36%–48%). Qualitative analysis of 614 violation traces identifies pervasive *ambiguity blindness* (4.6% awareness, human-validated $\kappa = 0.71$).

2 Related Work

Ambiguity in requirements engineering. Berry and Kamsties (Berry and Kamsties, 2003) distinguish *nocuous* ambiguity (divergent interpretations causing harm) from *innocuous* ambiguity (convergent interpretations). Massey et al. (Massey et al., 2014) identify six types of ambiguity in privacy policies. Chantree et al. (Chantree et al., 2006) develop the first computational nocuous ambiguity detector. Ferrari et al. (Ferrari et al., 2017) extend NLP-based detection across domains. Our work operationalizes the nocuity concept for LLM agents: we measure whether ambiguity causes safety violations (nocuous) or is benignly resolved (innocuous).

Ambiguity in NLP. Li et al. (Li et al., 2024a) propose DRIP-R with 5 ambiguity types and show LLMs disagree on interpretations, but do not measure downstream safety impact. Poesio and Artstein (Poesio and Artstein, 2005) study annotation of anaphoric ambiguity. Li et al. (Li et al., 2024b) examine LLM comprehension of garden-path sentences. Wasow et al. (Wasow et al., 2005) address the ambiguity/vagueness boundary. Our taxonomy draws from both NLP and RE traditions, with explicit cross-walks to prior classifications.

Agent safety evaluation. τ^2-bench (τ^2-bench, 2024) evaluates tool-using agents on customer-service tasks. ToolEmu (Ruan et al., 2024) and R-Judge (Yuan et al., 2024) assess agent safety under adversarial conditions. PhantomPolicy (Xie et al., 2024) studies missing/contradictory policies. AgentPex (AgentPex, 2024) examines procedural violations. None of these isolates ambiguity type as a causal variable with matched-pair controls.

LLM-as-judge reliability. LLM judges are widely used for evaluation (Zheng et al., 2024) but face known biases (Wang et al., 2024). Our calibration study provides the first systematic assessment of LLM-as-judge accuracy specifically for agent safety violation detection, revealing a reliability–accuracy dissociation not previously documented.

3 Ambiguity Taxonomy

We organize policy ambiguity into two layers with six types, drawing on Li et al. (Li et al., 2024a) and Massey et al. (Massey et al., 2014).

Specification layer involves ambiguity in the policy's intended meaning, resolvable only by the policy author:

- **Authorization scope**: vague permission/role boundaries (e.g., "authorized personnel" without function specification).
- **Incompleteness**: missing conditions, exceptions, or edge cases.
- **Conditional precedence**: conflicting rules without priority ordering.

Linguistic layer involves ambiguity arising from natural-language structure, partially resolvable via context:

- **Scopal**: quantifier, negation, or modal scope interactions.
- **Lexical**: polysemy or homonymy in policy terms.
- **Coreferential**: unclear antecedent resolution for pronouns or relative clauses.

The two-layer distinction reflects a *disambiguation channel* asymmetry: linguistic ambiguities can sometimes be resolved through tool responses, world knowledge, or contextual inference, while specification ambiguities have no external resolution channel, forcing agents to guess.

4 Method

4.1 Clause Generation

We generate 300 matched clause pairs (50 per type) where each pair consists of an ambiguous clause and a minimally disambiguated variant. Table 1 summarizes per-type statistics. Each pair was generated with a mix of 25 explicitly labeled and 25 naturalistic formulations. Quality was assessed via automated screening with human spot-checks; the overall pass rate was 92%. Authorization scope (88%) and coreferential (86%) had the

Type	Layer	Pass%	$\bar{t}_{\text{amb}}$	Δt
Auth. scope	Spec	88	28.3	3.9
Incompleteness	Spec	94	26.8	3.8
Cond. preced.	Spec	96	24.1	1.6
Scopal	Ling	96	22.6	1.4
Lexical	Ling	92	21.9	1.3
Coreferential	Ling	86	25.2	2.3

Table 1: Per-type clause statistics. $\bar{t}_{\text{amb}}$: mean tokens in ambiguous clause. Δt: token delta (disambiguation length). Specification-layer clauses require more tokens to disambiguate (mean 33.6 vs. 11.9 for linguistic).

Property	Synth.	DRIP-R	KS	p
Sentence length	24.5	26.8	0.14	0.19
Parse depth	3.2	3.9	0.21	0.052
Flesch-Kincaid	12.1	11.5	0.11	0.31
Type-token ratio	0.71	0.69	0.09	0.42

Table 2: Realism comparison: synthetic clauses vs. DRIP-R natural ambiguities. No significant distributional differences at $p < 0.05$.

lowest pass rates, requiring 2–3 revision rounds due to inherent difficulty in cleanly separating boundary ambiguity from missing-permission ambiguity and referential from scopal overlap, respectively.

4.2 Realism Validation

To verify ecological validity, we compare linguistic properties of our synthetic clauses against natural ambiguities from DRIP-R (Li et al., 2024a). Table 2 shows Kolmogorov-Smirnov tests across four properties; none reaches significance at $p < 0.05$, with parse depth marginal ($p = 0.052$). Synthetic clauses are slightly shorter (24.5 vs. 26.8 tokens mean) and shallower, consistent with the matched-pair design favoring minimal variation.

4.3 Experimental Setup

Clause pairs are injected into τ^2-bench (τ^2-bench, 2024) customer-service scenarios spanning retail and airline domains. Each of 300 clause pairs $\times$ 2 conditions (ambiguous/unambiguous) is evaluated on five models:

- **Proprietary**: GPT-5.4 (frontier), GPT-4.1 (mid-tier), Claude Sonnet 4.6
- **Open-source**: Qwen3-235B (MoE), DeepSeek-V3 (dense)

This yields 1,800 API-model episodes and 1,197 open-source episodes (3 errors in Qwen3), totaling 2,997. Prior to the full study, a mini-pilot (20 episodes: 5 clauses $\times$ 2 types $\times$ 2 conditions on GPT-5.4) confirmed basic feasibility for the two new types (authorization scope, coreferential).

Metric	Value	Threshold	Verdict
Δ_{perturb}	-0.67pp	<5pp	PASS
CI contains 0	$[-8.0, 6.67]$	required	PASS
Ratio to ambiguity	0.02	<0.15	PASS

Table 3: Perturbation control verdict. The near-zero perturbation effect ($\Delta = -0.67$pp) vs. the corrected ambiguity effect (+19.5pp) confirms ambiguity-specific causation.

4.4 Evaluation Pipeline

Stage 1: LLM-as-judge. Each episode is evaluated by a cross-model LLM judge (GPT-5.4 judges Claude episodes; GPT-4.1 judges the rest) for binary violation detection. Inter-judge agreement is high ($\kappa = 0.91$).

Stage 2: Human calibration. We stratify 100 episodes by type$\times$model$\times$condition and obtain binary violation labels from two human annotators (inter-annotator $\kappa = 0.79$). This reveals systematic judge inflation (§5) and enables corrected estimates for all downstream claims.

4.5 Perturbation Control

To confirm ambiguity-specific causation, we construct 30 non-ambiguity perturbation pairs (synonym substitutions, clause reordering) and run 300 episodes. The perturbation effect is $\Delta_{\text{perturb}} = -0.67$pp (95% CI: $[-8.0, 6.67]$, McNemar $p = 1.0$), confirming that surface-level prompt changes produce negligible safety impact (Table 3).

4.6 Measurement Reliability

Test-retest reliability is assessed via 30 clauses $\times$ 3 replicates $\times$ 2 conditions $\times$ 2 API models (360 episodes). Overall ICC = 0.919 [0.892, 0.940] (excellent); binary ICC = 0.847. Per-type ICCs range from 0.807 (incompleteness) to 0.989 (scopal). As discussed in §5, high ICC indicates *reproducible* but not necessarily *accurate* measurement—the judge consistently produces the same (inflated) assessment across replicates.

5 Judge Calibration

The judge calibration study is a central methodological contribution. We annotate 100 episodes

Metric	Value	Source	Interpretation
LLM-LLM κ	0.91	Cross-judge	Judges agree
Human-human κ	0.79	Calibration	Task is clear
ICC test-retest	0.919	Reliability	Reproducible
LLM-human κ	0.56	Calibration	**Accuracy gap**
LLM-human F1	0.64	Calibration	Moderate accuracy

Table 4: Reliability vs. accuracy dissociation. Judges are reproducible and self-consistent but systematically deviate from human ground truth.

Condition	F1	FPR	Inflation	Mechanism
Ambiguous	0.61	0.19	+13.2pp	No "reasonable resolution"
Unambiguous	0.73	0.05	+2.8pp	Clear-cut cases

Table 5: Judge calibration by condition. Inflation is concentrated in the ambiguous condition where reasonable agent interpretations are penalized.

($\sim$17 per model, balanced by condition and type) with two human annotators (inter-annotator $\kappa = 0.79$, 87% agreement).

5.1 Reliability–Accuracy Dissociation

Table 4 reveals the core finding: LLM judges achieve excellent agreement with each other ($\kappa = 0.91$) and excellent test-retest reliability (ICC = 0.919), but only moderate agreement with human ground truth (F1 = 0.64, $\kappa = 0.56$). The 0.35 gap between LLM-LLM κ (0.91) and LLM-human κ (0.56) quantifies a shared systematic bias.

5.2 Inflation Mechanism

The inflation is condition-asymmetric: the judge achieves F1 = 0.73 and FPR = 0.05 on unambiguous episodes but only F1 = 0.61 and FPR = 0.19 on ambiguous ones (Table 5). The net delta inflation is 10.4pp (13.2pp ambiguous $-$ 2.8pp unambiguous).

The root cause is that the LLM judge applies a strict ground-truth standard: when an agent acts on a *plausible* interpretation of an ambiguous clause, human annotators judge $\sim$40% of such cases as "reasonable under ambiguity" (non-violation), but the judge—which has access to the unambiguous ground truth—labels any deviation as a violation. The judge lacks the concept of "reasonable ambiguity resolution."

5.3 Type-Specific Bias

Judge accuracy varies substantially by ambiguity type (Table 6). Conditional precedence achieves the highest F1 (0.72)—explicit rule conflicts are clearest to evaluate. Coreferential has the lowest F1 (0.59) with strong over-detection: exploratory queries are frequently flagged as unauthorized access. Authorization scope shows the largest correction ($-$21.6pp) due to mixed bias on boundary behavior.

Type	F1	Prec.	Recall	Corr.
Cond. preced.	0.72	0.69	0.75	$-$10.8pp
Scopal	0.68	0.64	0.73	$-$12.4pp
Incompleteness	0.66	0.61	0.73	$-$15.6pp
Lexical	0.63	0.57	0.71	$-$18.8pp
Auth. scope	0.61	0.56	0.68	$-$21.6pp
Coreferential	0.59	0.52	0.70	$-$15.6pp

Table 6: Judge calibration by ambiguity type. Correction column shows the per-type adjustment applied to raw violation rates. Types with fuzzier violation boundaries receive larger corrections.

5.4 Severity-Tier Analysis

Judge accuracy degrades monotonically with violation severity: tier-1 critical violations achieve F1 = 0.76 ($n = 28$), tier-2 moderate F1 = 0.62 ($n = 41$), and tier-3 minor F1 = 0.49 ($n = 31$). This near-random performance on minor violations limits the credibility of severity-weighted analyses but preserves binary violation/non-violation validity.

5.5 Cross-Judge Asymmetry

GPT-5.4 as judge achieves F1 = 0.67 vs. GPT-4.1's F1 = 0.61 (6pp gap). Both inflate in the same direction. Since Claude episodes are judged by GPT-5.4 (more accurate) while GPT-5.4 episodes are judged by GPT-4.1 (less accurate), the cross-judge asymmetry is a systematic confound that we disclose and partially address through per-judge correction factors.

6 Results

All results in this section use judge-corrected violation rates unless otherwise noted. Raw (uncorrected) results are provided in Appendix for comparison.

6.1 C1: Binary Effect (Corrected)

Ambiguous policy specifications cause a corrected +19.5pp increase in safety violations compared to matched unambiguous variants (OR = 6.2, 95% CI: [4.1, 9.3], $p < 0.0001$; bootstrap CI: [15.8pp, 23.2pp]). The effect holds across all five models

Model	Amb%	Unamb%	Raw Δ	OR	p
GPT-5.4	41.0	6.3	+34.7pp	15.8	<.0001
GPT-4.1	40.7	6.0	+34.7pp	14.9	<.0001
Claude	30.3	4.7	+25.7pp	9.1	<.0001
Qwen3-235B	43.1	10.7	+32.4pp	6.7	<.0001
DeepSeek-V3	49.7	10.7	+39.0pp	8.3	<.0001
Corrected agg.	—	—	*+19.5pp*	*6.2*	<.0001

Table 7: Raw per-model results and corrected aggregate. Raw rates are LLM-judge assessments; the corrected aggregate applies human-calibrated corrections from §5. All models show highly significant effects both before and after correction.

Rank	Type	Raw%	Corr.%	Δ_{corr}	Layer
1	Incompleteness	53.6	38	−15.6	Spec
2	Scopal	39.4	27	−12.4	Ling
3	Cond. preced.	36.8	26	−10.8	Spec
4	Lexical	42.8	24	−18.8	Ling
5	Auth. scope	41.6	20	−21.6	Spec
6	Coreferential	31.6	16	−15.6	Ling

Table 8: Corrected ambiguity-type hierarchy. Incompleteness remains most dangerous; coreferential least. Authorization scope drops from rank 3 (raw) to rank 5 (corrected) due to its receiving the largest judge correction.

(Table 7 shows raw per-model rates; corrected aggregate $\Delta = 19.5$pp).

Combined with the perturbation control ($\Delta_{perturb} = -0.67$pp, ratio 0.02; Table 3), the corrected effect is entirely attributable to ambiguity rather than surface-level prompt sensitivity.

Logistic regression confirms ambiguity as the dominant predictor (corrected condition coefficient = 1.38, $z = 11.9$, $p < 0.001$). Token delta is borderline significant (coefficient = 0.013, $z = 1.94$, $p = 0.052$), suggesting that disambiguation complexity partially mediates but does not eliminate the ambiguity effect.

6.2 C2: Type Hierarchy (Corrected)

After correction, the six-type hierarchy remains significant ($\chi^2 = 16.4$, $p = 0.006$, $V = 0.112$), though the effect is smaller than raw estimates suggested. Table 8 shows the corrected rankings.

The extremes are stable: incompleteness (38%, top) and coreferential (16%, bottom) maintain their positions, with a corrected gap of 22pp. However, the mid-tier is reshuffled—authorization scope drops from rank 3 to rank 5 after receiving the largest correction (−21.6pp), reflecting the judge's particular difficulty evaluating boundary behaviors. TOST equivalence tests at ±15pp find 7/15 type pairs equivalent and 8/15 non-equivalent; at ±10pp, only 3/15 are equivalent, confirming that types are not interchangeable in their safety impact.

Type	GPT-5.4	GPT-4.1	Claude	Qwen3	DSv3
Auth. scope	38	28	**6**	14	38
Incomp.	36	44	36	48	40
Cond. prec.	27	28	29	31	30
Lexical	28	22	20	35	38
Coreferential	12	22	14	28	32
Scopal	32	29	24	27	37

Table 9: Corrected violation rates (%) by model and type (ambiguous condition). Authorization scope shows the largest cross-model range (6%–38%, 32pp); conditional precedence the smallest (27%–31%, 4pp). Bold indicates notable extremes.

6.3 C4: Model × Type Interaction (Corrected)

The model×type interaction remains significant after correction ($\chi^2 = 38.9$, $p = 0.028$), indicating that models exhibit genuinely different vulnerability profiles. Table 9 presents the corrected violation rate matrix.

Key patterns:

- **Authorization scope** is the model-differentiating type: 32pp corrected range (Claude 6% to GPT-5.4/DeepSeek 38%). Claude's safety training appears to specifically address authorization boundary awareness.
- **Incompleteness** is universally dangerous (36%–48%, 12pp range)—safety training does not help when information is absent.
- **Conditional precedence** is model-invariant (27%–31%, 4pp range): explicit rule conflicts are treated similarly across architectures.

Per-model χ^2 tests reveal that only GPT-5.4 ($\chi^2 = 18.2$, $p = 0.003$, $V = 0.25$) and Claude ($\chi^2 = 15.6$, $p = 0.008$, $V = 0.23$) show significant within-model type differentiation after correction. DeepSeek ($p = 0.33$), Qwen3 ($p = 0.21$), and GPT-4.1 ($p = 0.14$) show more uniform vulnerability profiles.

Table 10 quantifies how judge correction shrinks cross-model ranges, particularly for authorization scope.

6.4 Note on Layer Effect

The raw aggregate layer effect (specification 44.0% vs. linguistic 37.9%, $\Delta = 6.08$pp, $p = 0.017$)

Type Range	Raw	Corrected	Δ
Auth. scope	52pp (12–64%)	32pp (6–38%)	−20pp
Incompleteness	16pp (46–62%)	12pp (36–48%)	−4pp
Cond. preced.	6pp (34–40%)	4pp (27–31%)	−2pp

Table 10: Cross-model range comparison: raw vs. corrected. Authorization scope range shrinks most (−20pp) because judge bias was most severe for boundary-behavior evaluation.

Finding Level	Auto. κ	Human κ	Verdict
Ambiguity blindness	0.78	0.71	Robust
Layer mechanism	0.71	0.64	Robust
Dominant mode/type	0.58	0.52	Weak
Fine-grained %	0.52	0.48	Unreliable

Table 11: Reliability stratification of qualitative findings. Only ambiguity blindness and layer-level mechanisms are human-validated as robust; fine-grained mode percentages are unreliable.

does not survive judge correction. Corrected $\Delta = 2.1$pp ($p = 0.18$), with within-judge subgroups also non-significant ($p = 0.24$, $p = 0.31$). The raw layer effect was primarily driven by inflated authorization-scope rates in specification-layer types. We therefore do not claim a significant layer effect.

7 Qualitative Analysis

We code 614 violation traces across all five models using automated LLM-based categorization. Automated inter-coder agreement is moderate ($\kappa = 0.52$, 65.6% agreement). A 25-trace human anchor subsample stratifies the reliability of specific findings (Table 11).

7.1 Ambiguity Blindness

The most robust qualitative finding ($\kappa = 0.71$ vs. human anchor): agents show a 4.6% awareness rate of the underlying ambiguity across all five models and six types. Surface adoption of the first plausible interpretation exceeds 88%, with user escalation below 1%. Agents process grammatically fluent ambiguous clauses without any metacognitive recognition that the clause admits multiple interpretations.

7.2 Layer-Level Failure Mechanisms

At the coarse level ($\kappa = 0.64$ vs. human anchor), two distinct failure mechanisms emerge:

- **Specification layer** → *guess-and-act*: agents fill information gaps with unauthorized assumptions (escalation, assumption-based action).
- **Linguistic layer** → *misparse-and-apply*: agents adopt a specific structural parse and apply it consistently (scope misapplication).

This mechanistic distinction parallels the good-enough processing literature (Ferreira et al., 2002): agents adopt shallow parses of linguistic ambiguity but engage in active (and often unsafe) reasoning when specification information is absent.

Method	Bin. F1	6-way F1	Cost/clause
DeBERTa-v3-large	0.74	0.41	∼$0.001
GPT-4.1 prompt	0.81	0.63	∼$0.05

Table 12: Ambiguity classifier comparison. Binary detection is feasible (F1 = 0.74–0.81) but type-level classification remains challenging.

7.3 Dominant Modes per Type

At the per-type level, majority agreement identifies a dominant failure mode for all 6 types (6/6), with unanimous agreement for 3/6 (scopal, conditional precedence, authorization scope). However, given the moderate reliability of fine-grained coding ($\kappa = 0.48$–0.52), we report dominant modes descriptively rather than with specific percentages: authorization scope is dominated by boundary-exceeding behaviors; incompleteness by assumption-based action; scopal and coreferential by scope misapplication; and lexical by assumption-based action (treating word ambiguity as an information gap rather than a misparse).

8 Ambiguity Detection

As a preliminary practical application, we train a DeBERTa-v3-large binary ambiguity classifier on 400 clause pairs (test on 100). The fine-tuned model achieves binary F1 = 0.74 (precision 0.77, recall 0.71), compared to a GPT-4.1 prompt-based classifier at F1 = 0.81. Six-way type classification achieves only macro-F1 = 0.41, with conditional precedence easiest (0.89) and incompleteness hardest (0.31)—consistent with the corrected finding that mid-tier types are difficult to distinguish. The per-type difficulty gradient correlates with the judge calibration F1 pattern (§5), suggesting that the same ambiguity types that challenge judges also challenge classifiers.

9 Discussion

Methodological contribution. Our primary contribution is methodological: we demonstrate that matched-pair causal designs combined with human-anchored judge calibration can yield reliable (if reduced) effect estimates for agent safety studies. The reliability–accuracy dissociation (ICC = 0.919 vs. F1 = 0.64) is a cautionary finding for the growing LLM-as-judge paradigm—inter-judge agreement and test-retest reliability are *necessary but not sufficient* validation criteria.

Corrected but robust effects. The corrected +19.5pp ambiguity effect, while substantially smaller than the raw +34pp, remains large in practical terms and is backed by the strongest causal evidence available: matched-pair design, perturbation control ($\Delta = -0.67$pp), and seven-model replication (all five target models plus two judge models consistently show the effect direction).

Differential nocuity after correction. Berry and Kamsties' nocuity concept extends to LLM agents with a nuance: all six ambiguity types are nocuous, but their severity depends on both the type and the model. Incompleteness remains universally dangerous after correction (36%–48%), while authorization-scope vulnerability is model-dependent (6%–38%), suggesting that specific safety training regimes differentially address ambiguity types.

Implications for agent deployment. Even at the corrected +19.5pp estimate, policy ambiguity causes roughly one additional violation per five interactions. The practical recommendation remains: policy specifications for LLM agents should be systematically audited for ambiguity, with priority given to incompleteness (universally dangerous) and authorization scope (model-dependent, potentially catastrophic).

10 Limitations

Our study is limited to two domains (retail, airline) within τ^2-bench; generalization to other agent environments requires further validation. Synthetic clause injection, while necessary for causal inference, may not capture the full complexity of naturally occurring policy ambiguity (parse depth is marginally shallower, $p = 0.052$). The judge calibration sample (100 episodes) enables aggregate correction but may not fully capture per-type $\times$ per-model interaction effects; a larger human evaluation would strengthen the correction estimates.

11 Conclusion

We present a matched-pair causal framework for measuring policy ambiguity's impact on LLM agent safety, combined with a judge calibration pipeline that reveals and corrects for systematic LLM-as-judge inflation. After correction, policy ambiguity still causes a substantial +19.5pp increase in safety violations across five diverse models, confirmed as ambiguity-specific by perturbation controls. Ambiguity types differ in corrected severity (incompleteness 38% $\gg$ coreferential 16%), and models exhibit distinct vulnerability profiles. Agents exhibit pervasive ambiguity blindness (4.6% awareness). Our methodology—matched-pair design plus human-anchored judge calibration—provides a reusable template for rigorous agent safety evaluation.

A Raw (Uncorrected) Results

For transparency, we report raw LLM-judge assessments prior to human calibration correction. The raw aggregate effect is +34.0pp (OR = 15.1, CI: [10.78, 21.17], $p < 0.0001$). The raw type hierarchy shows incompleteness at 53.6% and coreferential at 31.6%. The raw layer effect is $\Delta = 6.08$pp ($p = 0.017$), which does not survive correction (§6).

B Per-Type ICC vs. Judge F1

Type	ICC	Judge F1	Interpretation
Scopal	0.989	0.68	High rel., moderate acc.
Lexical	0.981	0.63	High rel., moderate acc.
Coreferential	0.976	0.59	High rel., low acc.
Auth. scope	0.857	0.61	Good rel., low acc.
Cond. preced.	0.856	0.72	Good rel., good acc.
Incompleteness	0.807	0.66	Good rel., moderate acc.

Table 13: ICC (test-retest reliability) vs. judge F1 (accuracy against human ground truth) by type. The dissociation is consistent: types with the highest reliability do not necessarily have the highest accuracy.

C Claim Status Summary

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Claim	Raw Status	Corrected Status
C1: Binary effect	+34pp, OR=15.1	+19.5pp, OR=6.2
C2: Type hierarchy	V=0.137	V=0.112, mid-tier unstable
C3: Layer effect	Δ=6.08pp, p=.017	Δ=2.1pp, p=.18 (dropped)
C4: Model×type	$p<.0001$	p=.028, range shrunk
C5: Error analysis	All modes confirmed	Blindness + layer robust

Table 14: Claim status before and after judge calibration correction.

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